

Exercise 8-1: Schedule of Expected Cash Collections

Given:

Peak sales for Midwest Products, a wholesale distributor of leaf rakes, occur in August. The company's sales budget for the third quarter showing these peak sales is given below:

	May	June	July	August	September
Budgeted Sales (All on Account)			\$600,000	\$900,000	\$500,000
Actual Sales (All on Account)	\$430,000	\$540,000			
Total Sales	\$430,000	\$540,000	\$600,000	\$900,000	\$500,000

From past experience, the company has learned that 20% of a month's sales are collected in the month of sale, another 70% are collected in the month following the sale, and the remaining 10% are collected in the second month following the sale. Bad debts are negligible and can be ignored.

Required:

1. Prepare a schedule of expected cash collections from sales, by month and in total, for the third quarter.

	May	June	July	August	September
20% collected in month of sale	\$86,000	\$108,000	\$120,000	\$180,000	\$100,000
70% collected in month after sale		301,000	378,000	420,000	630,000
10% collected in 2nd month after			43,000	54,000	60,000
Total	\$86,000	\$409,000	\$541,000	\$654,000	\$790,000

2. Assume that the company will prepare a budgeted balance sheet as of September 30. Compute the A/R as of that date.

Accounts Receivable – September 30:		%	
		Uncollected	A/R
From August Sales:	\$900,000	10%	\$90,000
From September Sales:	\$500,000	80%	400,000
Total A/R			\$490,000

Check:

Total Sales:	\$2,970,000
Less Total Collections:	2,480,000
Accounts Receivable -- September 30:	\$490,000

Total
\$2,000,000
\$2,970,000

3

Total
\$400,000
1,428,000
157,000
\$1,985,000

Exercise 8-2: Production Budget

Given:

Crystal telecom has budgeted the sales of its innovative mobile phone over the next four months as follows:

	Unit Sales
July	30,000
August	45,000
September	60,000
October	50,000

The company is now in the process of preparing a production budget for the third quarter. Past experience has shown that end-of-month inventories of finished goods must equal 10% of the next month's sales. The inventory at the end of June was 3,000.

Required:

Prepare a production budget for the third quarter showing the number of units to be produced each month and for the quarter in total.

	July	August	September	Quarter	October
Budgeted Sales	30,000	45,000	60,000	135,000	50,000
Desired EI**	4,500	6,000	5,000	5,000	
Total Units Desired	34,500	51,000	65,000	140,000	
Less BI	3,000	4,500	6,000	3,000	
Production	31,500	46,500	59,000	137,000	
				137,000	

** EI is anticipated to be 10% of the next month's sales

Exercise 8-3: Direct Materials Budget

Given:

Micro Products, Inc. has developed a very powerful electronic calculator. Each calculator requires 3 small "chips" that cost \$2 each and are purchased from an overseas supplier. Micro Products has prepared a production budget for the calculator by quarters for Year 2 and for the first quarter of Year 3, as shown below.

	1st Quarter Y2	2nd Quarter Y2	3rd Quarter Y2	4th Quarter Y2
Budgeted Production (Calculators)	60,000	90,000	150,000	100,000

The chip used in production of the calculator is sometimes hard to get, so it is necessary to carry large inventory precaution against stockout. For this reason, the inventory of chips at the end of a quarter must be equal to the following quarter's production needs. Some 36,000 chips will be on hand to start the first quarter of Year 2.

Required:

Prepare a direct materials budget for chips, by quarter and in total, for Year 2. At the bottom of your budget show the dollar amount of purchases for each quarter and for the year in total.

	1st Quarter Y2	2nd Quarter Y2	3rd Quarter Y2	4th Quarter Y2
Budgeted Production (Calculators)	60,000	90,000	150,000	100,000
# of chips needed per calculator	3	3	3	3
Total chips needed for production	180,000	270,000	450,000	300,000
Desired EI **	54,000	90,000	60,000	48,000
Total direct materials needed	234,000	360,000	510,000	348,000
Less: BI of chips	36,000	54,000	90,000	60,000
Required purchases of chips (units)	198,000	306,000	420,000	288,000
Cost per chip	\$2	\$2	\$2	\$2
Required purchases of chips (dollars)	\$396,000	\$612,000	\$840,000	\$576,000

** EI is anticipated to be 20% of the next quarter's production needs.

W:

1st Quarter Y3
80,000

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Year 2
<u>400,000</u>
<u>3</u>
1,200,000
<u>48,000</u>
1,248,000
<u>36,000</u>
1,212,000
<u>\$2</u>
<u>\$2,424,000</u>

Exercise 8-4: Direct Labor Budget**Given:**

The Production Department of the Riverside Plant of Junnen Corporation has submitted the following forecast of units to be produced at the plant for each quarter of the upcoming fiscal year. The plant produces high-end outdoor barbecue grills.

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
Units to be produced	5,000	4,400	4,500	4,900

Each unit requires 0.40 direct labor hours and direct labor workers are paid \$11 per hour.

Required:

- Construct the company's direct labor budget for the upcoming fiscal year, assuming that the direct labor workforce is adjusted each quarter to match the number of hours required to produce the forecasted number of units produced.

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	Year
Units to be produced	5,000	4,400	4,500	4,900	18,800
Direct labor hours required per grill	0.4	0.4	0.4	0.4	0.4
Total DLHs required for production	2,000	1,760	1,800	1,960	7,520
DL Cost per hour	\$11	\$11	\$11	\$11	\$11
Total Budgeted DL Costs	\$22,000	\$19,360	\$19,800	\$21,560	\$82,720

- Construct the company's direct labor budget for the upcoming fiscal year, assuming that the direct labor workforce is not adjusted each quarter. Instead assume that the company's direct labor workforce consists of permanent employees who are guaranteed to be paid for at least 1,800 hours of work each quarter. If the number of required direct labor hours is less than this number, the workers are paid for 1,800 hours anyway. Any hours worked in excess of 1,800 hours in a quarter are paid at the rate of 1.5 times the normal hourly rate for DL.

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	Year
Units to be produced	5,000	4,400	4,500	4,900	18,800
Direct labor hours required per grill	0.40	0.40	0.40	0.40	0.40
Total DLHs required for production	2,000	1,760	1,800	1,960	7,520
Minimum guaranteed hours (@ \$11)	1,800	1,800	1,800	1,800	7,200
Wage rate for guaranteed hours	\$11	\$11	\$11	\$11	\$11
Wages for guaranteed time	\$19,800	\$19,800	\$19,800	\$19,800	\$79,200
Overtime hours	200	0	0	160	360
Wages rate for overtime hours (1.5X)	\$16.50	\$16.50	\$16.50	\$16.50	\$16.50
Wages for overtime	\$3,300	\$0	\$0	\$2,640	\$5,940
Total Budgeted DL Costs	\$23,100	\$19,800	\$19,800	\$22,440	\$85,140

Exercise 8-5: Manufacturing Overhead Budget**Given:**

The direct labor budget of Krispin Corporation for the upcoming fiscal year contains the following details concerning budgeted direct labor hours.

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
Budgeted DL Hours	5,000	4,800	5,200	5,400

The company's VMOH rate is **\$1.75** per DLH and the company's FMOH is **\$35,000** per quarter. The only noncash item included in the FMOH is depreciation, which is \$15,000 per quarter.

Required:

1. Construct the company's MOH budget for the upcoming fiscal year.

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
Budgeted DL Hours	5,000	4,800	5,200	5,400
VMOH rate per DLH	\$1.75	\$1.75	\$1.75	\$1.75
Budgeted VMOH Expense	\$8,750	\$8,400	\$9,100	\$9,450
Budgeted FMOH Expense	35,000	35,000	35,000	35,000
Budgeted TMOH Expense	\$43,750	\$43,400	\$44,100	\$44,450

2. Compute the company's TMOH rate for the upcoming fiscal year. \$8.6127

Year
20,400
\$1.75
\$35,700
140,000
\$175,700

Exercise 8-6: Selling and Administrative Expense Budget**Given:**

The budgeted unit sales of Haerve Company for the upcoming fiscal year are provided below:

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
Budgeted unit sales	12,000	14,000	11,000	10,000

The company's variable selling and administrative expense per unit is **\$2.75**. Fixed selling and Administrative expenses include advertising expenses of **\$12,000** per quarter, executive salaries of **\$40,000** per quarter and depreciation of **\$16,000** per quarter. In addition, the company will make insurance payments of **\$6,000** in the 2nd Quarter and **\$6,000** in the 4th Quarter. Finally, property taxes of **\$6,000** will be paid in the 3rd Quarter.

Required:

Prepare the company's selling and administrative expense budget for the upcoming fiscal year.

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
Budgeted units	12,000	14,000	11,000	10,000
Variable S & A rate per unit	\$2.75	\$2.75	\$2.75	\$2.75
Budgeted Variable S & A Expense	\$33,000	\$38,500	\$30,250	\$27,500
Budgeted Fixed S & A Expenses:				
Advertising	\$12,000	\$12,000	\$12,000	\$12,000
Executive salaries	40,000	40,000	40,000	40,000
Depreciation	16,000	16,000	16,000	16,000
Insurance Expense ¹		6,000		6,000
Property Tax Expense ¹			6,000	
Total Fixed S & A Expense	\$68,000	\$74,000	\$74,000	\$74,000
Total Budgeted S&A Expense	\$101,000	\$112,500	\$104,250	\$101,500

¹ Are these expense items or cash payments?

ative

100

Year
47,000
\$2.75
\$129,250
\$48,000
160,000
64,000
12,000
6,000
\$290,000
\$419,250

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Problem 9-9 (45 minutes)

1. Schedule of expected cash collections:

	<i>Month</i>			
	<i>July</i>	<i>August</i>	<i>September</i>	<i>Quarter</i>
From accounts receivable:				
May sales				
\$250,000 × 3%	\$ 7,500			\$ 7,500
June sales				
\$300,000 × 70%	210,000			210,000
\$300,000 × 3%		\$ 9,000		9,000
From budgeted sales:				
July sales				
\$400,000 × 25%	100,000			100,000
\$400,000 × 70%		280,000		280,000
\$400,000 × 3%			\$ 12,000	12,000
August sales				
\$600,000 × 25%		150,000		150,000
\$600,000 × 70%			420,000	420,000
September sales				
\$320,000 × 25%			80,000	80,000
Total cash collections.....	<u>\$317,500</u>	<u>\$439,000</u>	<u>\$512,000</u>	<u>\$1,268,500</u>

Problem 9-9 (continued)

2. Cash budget:

	<i>Month</i>			
	<i>July</i>	<i>August</i>	<i>September</i>	<i>Quarter</i>
Cash balance, beginning	\$ 44,500	\$ 28,000	\$ 23,000	\$ 44,500
Add receipts:				
Collections from customers.....	<u>317,500</u>	<u>439,000</u>	<u>512,000</u>	<u>1,268,500</u>
Total cash available.....	<u>362,000</u>	<u>467,000</u>	<u>535,000</u>	<u>1,313,000</u>
Less disbursements:				
Merchandise purchases ...	180,000	240,000	350,000	770,000
Salaries and wages	45,000	50,000	40,000	135,000
Advertising	130,000	145,000	80,000	355,000
Rent payments	9,000	9,000	9,000	27,000
Equipment purchases	<u>10,000</u>	<u>—</u>	<u>—</u>	<u>10,000</u>
Total disbursements.....	<u>374,000</u>	<u>444,000</u>	<u>479,000</u>	<u>1,297,000</u>
Excess (deficiency) of receipts over disbursements	<u>(12,000)</u>	<u>23,000</u>	<u>56,000</u>	<u>16,000</u>
Financing:				
Borrowings	40,000	—	—	40,000
Repayments	—	—	(40,000)	(40,000)
Interest	—	—	(1,200)	(1,200)
Total financing	<u>40,000</u>	<u>—</u>	<u>(41,200)</u>	<u>(1,200)</u>
Cash balance, ending.....	<u>\$ 28,000</u>	<u>\$ 23,000</u>	<u>\$ 14,800</u>	<u>\$ 14,800</u>

3. If the company needs a \$20,000 minimum cash balance to start each month, then the loan cannot be repaid in full by September 30. If the loan is repaid in full, the cash balance will drop to only \$14,800 on September 30, as shown above. Some portion of the loan balance will have to be carried over to October, at which time the cash inflow should be sufficient to complete repayment.

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Problem 9-12 (30 minutes)

1.

Zan Corporation
Direct Materials Budget

	<i>1st Quarter</i>	<i>2nd Quarter</i>	<i>3rd Quarter</i>	<i>4th Quarter</i>	<i>Year</i>
Required production (units)	5,000	8,000	7,000	6,000	26,000
Raw materials per unit (grams)	<u>× 8</u>	<u>× 8</u>	<u>× 8</u>	<u>× 8</u>	<u>× 8</u>
Production needs (grams)	40,000	64,000	56,000	48,000	208,000
Add desired ending inventory (grams)	<u>16,000</u>	<u>14,000</u>	<u>12,000</u>	<u>8,000</u>	<u>8,000</u>
Total needs (grams)	56,000	78,000	68,000	56,000	216,000
Less beginning inventory (grams)	<u>6,000</u>	<u>16,000</u>	<u>14,000</u>	<u>12,000</u>	<u>6,000</u>
Raw materials to be purchased (grams)	<u>50,000</u>	<u>62,000</u>	<u>54,000</u>	<u>44,000</u>	<u>210,000</u>
Cost of raw materials to be purchased at \$1.20 per gram	<u>\$60,000</u>	<u>\$74,400</u>	<u>\$64,800</u>	<u>\$52,800</u>	<u>\$252,000</u>

Schedule of Expected Cash Disbursements for Materials

Accounts payable, beginning balance	\$ 2,880				\$ 2,880
1st Quarter purchases	36,000	\$24,000			60,000
2nd Quarter purchases		44,640	\$29,760		74,400
3rd Quarter purchases			38,880	\$25,920	64,800
4th Quarter purchases				31,680	31,680
Total cash disbursements for materials	<u>\$38,880</u>	<u>\$68,640</u>	<u>\$68,640</u>	<u>\$57,600</u>	<u>\$233,760</u>

Problem 9-12 (continued)

2.

Zan Corporation
Direct Labor Budget

	<i>1st Quarter</i>	<i>2nd Quarter</i>	<i>3rd Quarter</i>	<i>4th Quarter</i>	<i>Year</i>
Required production (units)	5,000	8,000	7,000	6,000	26,000
Direct labor-hours per unit	<u>× 0.20</u>	<u>× 0.20</u>	<u>× 0.20</u>	<u>× 0.20</u>	<u>× 0.20</u>
Total direct labor-hours needed	1,000	1,600	1,400	1,200	5,200
Direct labor cost per hour	<u>× \$11.50</u>	<u>× \$11.50</u>	<u>× \$11.50</u>	<u>× \$11.50</u>	<u>× \$11.50</u>
Total direct labor cost	<u>\$ 11,500</u>	<u>\$ 18,400</u>	<u>\$ 16,100</u>	<u>\$ 13,800</u>	<u>\$ 59,800</u>

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Problem 9-11 (45 minutes)

	<i>July</i>	<i>August</i>	<i>Septem-ber</i>	<i>October</i>
1. Production budget:				
Budgeted sales (units).....	35,000	40,000	50,000	30,000
Add desired ending inventory...	<u>11,000</u>	<u>13,000</u>	<u>9,000</u>	<u>7,000</u>
Total needs.....	46,000	53,000	59,000	37,000
Less beginning inventory	<u>10,000</u>	<u>11,000</u>	<u>13,000</u>	<u>9,000</u>
Required production.....	<u>36,000</u>	<u>42,000</u>	<u>46,000</u>	<u>28,000</u>

2. During July and August the company is building inventories in anticipation of peak sales in September. Therefore, production exceeds sales during these months. In September and October inventories are being reduced in anticipation of a decrease in sales during the last months of the year. Therefore, production is less than sales during these months to cut back on inventory levels.

3. Raw direct materials budget:

	<i>July</i>	<i>August</i>	<i>Sep-tember</i>	<i>Third Quarter</i>
Required production (units)	36,000	42,000	46,000	124,000
Material H300 needed per unit.....	<u>× 3 cc</u>	<u>× 3 cc</u>	<u>× 3 cc</u>	<u>× 3 cc</u>
Production needs (cc)	108,000	126,000	138,000	372,000
Add desired ending inventory (cc).....	<u>63,000</u>	<u>69,000</u>	<u>42,000</u>	* <u>42,000</u>
Total material H300 needs	171,000	195,000	180,000	414,000
Less beginning inventory (cc)...	<u>54,000</u>	<u>63,000</u>	<u>69,000</u>	<u>54,000</u>
Material H300 purchases (cc)...	<u>117,000</u>	<u>132,000</u>	<u>111,000</u>	<u>360,000</u>

* 28,000 units (October production) × 3 cc per unit = 84,000 cc;
84,000 cc × 1/2 = 42,000 cc.

As shown in part (1), production is greatest in September; however, as shown in the raw direct materials budget, purchases of materials are greatest a month earlier—in August. The reason for the large purchases of materials in August is that the materials must be on hand to support the heavy production scheduled for September.